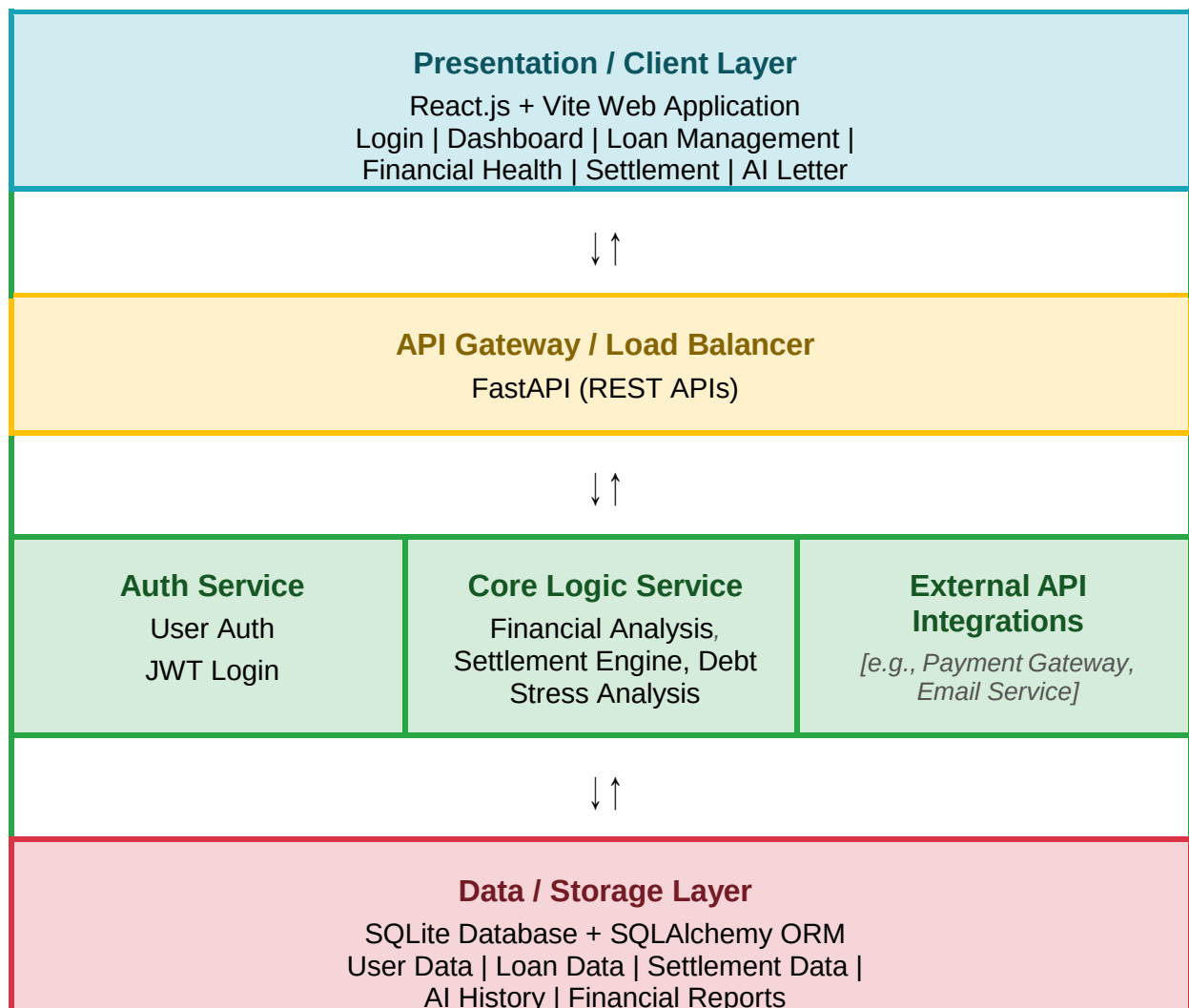


Solution Architecture

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive user interface for registration, login, loan management, financial analysis, settlement recommendations, dashboard, and AI negotiation letters.	React.js, Vite, HTML5, CSS3, JavaScript
API Gateway	Handles all client requests, routes APIs to backend services, validates requests, and manages communication between frontend and backend.	FastAPI, REST API
Microservices / Core Logic	Performs authentication, loan management, financial health analysis, debt stress calculation, settlement prediction, and AI negotiation letter generation.	Python 3.11, FastAPI, SQLAlchemy
Database	Stores user profiles, loan records, financial analysis results, settlement history, and AI-generated negotiation history securely.	SQLite, SQLAlchemy ORM